

Statistical Analysis

VOSviewer, a bibliometric visualization program intended to analyze the intellectual connections among scientific articles, was used to conduct the bibliographic coupling analysis. Documents were chosen as the unit of analysis after the bibliographic records from the scopus dataset were exported from the Scopus database and loaded into VOSviewer. Each common reference between two publications might contribute equally to the overall coupling strength when the Full Counting approach is used.

Bibliographic coupling looks at the references mentioned by the publications themselves, as opposed to co-citation analysis, which quantifies how often two publications are cited together by later research. Because they rely on similar theoretical frameworks, methodological approaches, and fundamental evidence, studies with a higher percentage of references are thought to have stronger intellectual ties. Because of this, bibliographic coupling is especially useful for spotting current research trends and figuring out the connections between recent works.

The bibliographic coupling intensity between each pair of publications in the dataset was determined by VOSviewer. Each publication is represented by a node in the network, and the number of references shared between studies determines the strength of the bibliographic relationship, which is represented by the connecting lines. Greater integration within the research network is reflected in larger nodes, which represent publications with stronger overall linkages to the larger body of literature. Stronger bibliographic ties are also indicated by thicker connecting lines, which imply that the linked studies are based on a very comparable corpus of prior research. VOSviewer's distance-based visualization technique was used to create the spatial layout of nodes; publications that are closer together have more similar reference lists than those that are farther away.

The network was then divided into colour-coded groups by the program using a clustering process. Every cluster is a group of papers that look into linked scientific issues or research themes since they have many references in common. By identifying groups of articles that have grown around shared theoretical ideas, methodological approaches, or clinical uses, these clusters offer insight into the intellectual organization of the literature. As a result, the resulting image provides a conceptual and quantitative summary of the evolution of the field's study.

Figure Significance

Research on brain-computer interface (BCI) technology has grown into a highly interconnected and multidisciplinary body of literature, as the bibliographic coupling network shows. The included studies are arranged into multiple distinct but overlapping research clusters rather than existing as isolated investigations, demonstrating that researchers often build upon a common foundation of scientific evidence while investigating various neurological conditions and rehabilitation strategies. Motor rehabilitation after stroke and spinal cord damage is one of the network's most significant issues. In order to enhance motor recovery, studies in this cluster

examine how BCIs can be integrated with technologies including robotic rehabilitation devices, functional electrical stimulation (FES), and gait-training systems. When creating rehabilitation interventions, researchers frequently depend on well-established ideas of neuroplasticity, motor learning, and cortical reorganization, according to the robust bibliographic linkages found among these papers. This reflects a growing understanding that the best results from BCI-assisted rehabilitation come from combining repetitive, task-specific motor training with neural decoding technologies.

Research on cortical plasticity and motor imagery is highlighted in a second cluster. These studies explore the extent to which repeated motor imagery training can aid in functional recovery and look at how imagined movement stimulates sensorimotor brain regions despite severe neurological impairment. Research on motor imagery frequently draws on related basic work in neuroscience, especially research examining cortical remapping, sensorimotor representation, and activity-dependent neuroplasticity, according to the close coupling discovered within this cluster. These references' consistency shows that motor imagery has established itself as a key element of modern BCI rehabilitation research. Studies on pain perception and neurofeedback-based therapies make up a third research grouping. In order to better understand and manage chronic pain problems, such as phantom limb pain and neuropathic pain after neurological injury, these works examine the use of EEG signals, immersive virtual reality, and neurofeedback systems. Despite the fact that these studies focus on different clinical populations than the motor rehabilitation studies, their close proximity within the network suggests that they still use many of the same neuroscientific concepts, especially those related to sensorimotor cortical activity, neural adaptation, and the brain's functional organization. Crucially, the overlap between these clusters shows how breakthroughs in one field of BCI research often influence those in another. For instance, concepts that were first created to enhance motor rehabilitation are increasingly being used to treat pain, and research on brain plasticity has impacted both prosthetic control and rehabilitation.

Connection to Included Articles

The bibliographic coupling analysis demonstrates the intellectual connections between the eleven articles in this review and represents the main research themes found throughout. The research constantly rely on common theoretical underpinnings involving neuroplasticity, motor imagery, brain decoding, and adaptive rehabilitation technology, despite the fact that they examine various neurological illnesses and rehabilitation goals. Brain-computer interface-assisted motor rehabilitation is the subject of a significant amount of the literature. Effects of Brain-Computer Interface-Controlled Functional Electrical Stimulation Training on Shoulder Subluxation for Patients with Stroke, for instance, assesses how well BCI devices and functional electrical stimulation work together to enhance upper-limb rehabilitation after stroke. Similar to this, Brain-Computer Interface-Based Robotic End Effector System for Wrist and Hand Rehabilitation studies robotic-assisted therapy for chronic stroke patients, while The Promotoer uses

BCI-assisted rehabilitation protocols to evaluate early and long-term motor recovery. These research studies consistently highlight the significance of activity-dependent neuroplasticity and motor relearning, which explains their strong bibliographic links within the network despite variations in rehabilitation approaches and patient demographics.

Rehabilitating people with spinal cord injuries using brain-machine interfaces and motor imagery is another well-known cluster. Long-Term Training with a Brain-Machine Interface-Based Gait Protocol Induces Partial Neurological Recovery in Paraplegic Patients shows that people with persistent paraplegia can have quantifiable neurological improvements through extended BCI-assisted gait training. This result is further supported by Training with Brain-Machine Interfaces, Visuotactile Feedback, and Assisted Locomotion Improves Sensorimotor, Visceral, and Psychological Signs in Chronic Paraplegic Patients, which shows how repeated multimodal rehabilitation can enhance both motor and non-motor outcomes. Furthermore, Motor Imagery in Spinal Cord Injured People is Modulated by Somatotopic Coding, Perspective Taking, and Post-Lesional Chronic Pain explores the brain mechanisms that underlie motor imagery after spinal cord injury, offering crucial theoretical backing for the rehabilitation strategies assessed in the intervention studies. When taken as a whole, these works show how clinical rehabilitation research and experimental neuroscience are complementary, leading to a strong bibliographic linkage. A significant corpus of research on pain perception and neurofeedback treatments is also identified by the study. Research like EEG-Based Brain-Computer Interface with Immersive Virtual Reality for Phantom Limb, Neurofeedback Training without Explicit Phantom Hand Movements and Hand-Like Visual Feedback to Modulate Pain presents novel methods for tracking and altering brain activity linked to persistent pain. By showing how brain decoding and real-time feedback may enhance pain management and quality of life for people with neurological illnesses, these research jointly show the growing relevance of BCI technology beyond motor rehabilitation. Their bibliographic relationships imply that while extending these ideas to a wider range of clinical applications, researchers are still building on comparable findings on brain plasticity, EEG signal processing, and sensorimotor representation.

A number of general ideas, such as neuroplasticity, cortical rearrangement, motor imagery, and adaptive rehabilitation, consistently surface in all of the included investigations. Despite variations in study design, participant populations, and rehabilitation outcomes, the bibliographic coupling network demonstrates how these ideas function as shared theoretical underpinnings. This pattern indicates that the discipline is gradually moving toward an integrated concept of neurological rehabilitation, where discoveries in one area like motor recovery inform advancements in related areas, such as pain modulation, prosthesis control, and functional restoration. All things considered, the bibliographic coupling analysis offers strong proof that modern BCI research is distinguished by a high degree of logical coherence. Researchers are simultaneously expanding the application of a common body of knowledge across various neurological disorders, as seen by the shared citation patterns found throughout the network.

Because of this, the figure not only shows how the literature is currently organized, but it also emphasizes the collaborative and interdisciplinary trajectory that will probably influence future advancements in brain-computer interface research.

Methodology and Code

VOSviewer (version X.X) was used to create the bibliographic coupling network in order to analyze the intellectual connections between the research that were part of this review. Using pre-established search phrases associated with brain-computer interfaces and neurological rehabilitation, bibliographic records were initially obtained from the Scopus database. The completed dataset was generated in CSV format and imported into VOSviewer for analysis following the application of the inclusion and exclusion criteria. Documents were chosen as the analytical unit and bibliographic coupling as the analytical technique within VOSviewer. In order to ensure that every common reference contributed equally to the coupling strength between publications, the Full Counting technique was used. In order to create a network visualization, VOSviewer determined how many references each pair of included studies exchanged. The software's clustering technique aggregated connected studies into color-coded clusters that represented different research themes, and publications with stronger bibliographic ties were placed closer together within the network. While the thickness of connecting lines indicates the intensity of the bibliographic coupling between studies, node size indicates the relative strength of each publication's links within the network. By identifying groups of studies that rely on comparable theoretical frameworks and fundamental evidence, this methodology makes it possible to identify the intellectual structure of the literature. The resulting network sheds light on the evolution of brain-computer interface research in several neurological rehabilitation domains, such as pain management, motor imagery, neuroplasticity, stroke recovery, and spinal cord injury.

Code

VOSviewer was used to create the bibliometric analysis and network visualization. The typical VOSviewer approach for bibliographic coupling analysis was followed for data preprocessing and figure creation. The accompanying GitHub repository contains any extra preparation or data management done in R or Bibliometrix.